MeLALight: Melanoma Classification Using Multi-Stage DCT-Based Frequency-Channel Attention Fusion on RepViT

***Abstract*—Melanoma is the most lethal type of skin cancer, being responsible for most skin-cancer deaths despite its rare occurrence among diagnoses, as survival is closely associated with detection time. Unfortunately, benign and malignant lesions have similar dermoscopic appearance, making both interpretation and automated screening challenging tasks. In this paper, we introduce MeLALight, an efficient classifier model with DCT-based FcaNet frequency-channel attention applied to the last two stages of a pretrained RepViT-M0.9 backbone and then feature recalibration followed by fusing and processing through lightweight MLP classification head. On a dataset of 10,605 dermoscopic images with stratified five-fold cross-validation and sealed holdout of 1,000 images, MeLALight achieves average validation AUC of 0.9817 ($\pm$0.0026); holdout AUC is 0.9772, accuracy 91.80%, precision 95.24%, specificity 95.60%, and F1 score 91.48%. The total number of parameters is 4.95M, and number of FLOPs is 1.6335B. Ablation analysis shows that attention at the last two stages provides the optimal combination of accuracy, stability over folds and memory footprint, making MeLALight a promising solution for computer-aided melanoma screening.**

***Index Terms*—Melanoma, RepViT, FcaNet, Frequency-Channel Attention, Feature Fusion, Efficient Deep Learning**

I. Introduction

Skin cancer remains one of the most common malignancies worldwide. Melanoma, the malignant proliferation of melanocytes, is the deadliest form and accounts for the greatest number of skin-cancer deaths despite its relatively low incidence. Five-year survival drops considerably with disease progression, making early detection critical. Dermoscopy improves early diagnosis, yet benign and malignant lesions frequently share similar color, texture, and border characteristics, complicating both clinical interpretation and automated screening.

Deep learning has advanced computer-aided skin lesion diagnosis, but many state-of-the-art models rely on large backbones such as DenseNet-121 or ResNet-50, or on computationally expensive ensembles [1], [2]. Standard channel attention mechanisms like Squeeze-and-Excitation (SE) [3] use global average pooling exclusively, capturing only the lowest-frequency spatial information while discarding higher-frequency content. FcaNet [4] addresses this limitation by incorporating the Discrete Cosine Transform (DCT) [5] into channel attention computation. However, combining FcaNet with modern mobile-grade architectures such as RepViT [6] for skin lesion analysis remains unexplored.

We propose MeLALight, which integrates three complementary design choices:

1) A 4.95M-parameter melanoma classifier combining a mobile-grade RepViT backbone with multi-stage DCT-based FcaNet frequency-channel attention applied to its two deepest stages.
2) A multi-stage fusion strategy that recalibrates selected hierarchical features before concatenation, enabling the classification head to leverage both texture and semantic cues.
3) A rigorous evaluation pipeline comprising stratified five-fold cross-validation, a sealed 1,000-image holdout set, and comprehensive computational profiling.

II. Related Work

A. Melanoma Classification on MSCD-10K

The publicly available Melanoma Skin Cancer Dataset of 10,000 images (MSCD-10K) [7] dataset is one of the widely used datasets to benchmark melanoma classification, where the existing literature includes several well-known families of architectures.

Sabir and Mehmood [8] have conducted an experiment comparing a simple CNN, ResNet-18, and EfficientNet-B0 trained with the same transfer learning settings. The performance of EfficientNet-B0 was 97% accuracy, proving the importance of pretraining of features.

Ensemble methods dominate the accuracy leaderboard. Ghosh et al. [9] pooled embeddings from a DCNN, a CapsNet, and a ViT, then applied a majority vote across Random Forest, XGBoost, SVM, KNN, and logistic regression, reaching 91.6% accuracy. Albashish et al. [2] hard-voted three fine-tuned networks—VGG16, NASNetMobile, and ResNet50V2—reporting 91.50% in the abstract (90.86% $\pm$ 0.42 in the body). This discrepancy highlights the importance of transparent ensemble reporting.

Hybrid architectures take an intermediate place among all of the other types. In particular, Koteswara Rao et al. [10] have combined the Vision Transformer with ResNet-50, thus combining both local and global feature analysis (94.00%). Moreover, Rao et al. [11] have applied another method of training, where CNN features were processed using LSTM to find sequential dependencies, leading to 93.40% accuracy.

There have been multiple studies where good results were achieved with rather small backbone networks. Three-layered

convolutional neural network by Lankadasu et al. [12] achieves above 95% accuracy on almost 10,000 publicly available images. The DeepMelaNet, proposed by Al Huda et al. [13], which is a more compact version of EfficientNet, reaches 93.40% accuracy.

The preprocessing of images can be another important component in achieving high results. Quishpe-Usca et al. [14] combined morphological hair removal with wavelet filtering, lifting AlexNet to 91.30%. Yaqoob et al. [15] merged MSCD-10K with SIIM-ISIC before fine-tuning VGG-16, obtaining 89.00%—a reminder that external data changes the task.

The highest accuracies have been obtained using SmartSkin-XAI [1] (DenseNet-121 + Grad-CAM, 98.00%) and Balaha et al. [16] (U-Net segmentation followed by HHO-tuned CNNs, 97.08%).

In all these cases, there is one clear trend that stands out: accuracy for the MSCD-10K dataset [7] increases with the size and number of parameters of the network used. Ensembles of networks as well as networks having more than 100 million parameters have dominated the leaderboard so far, whereas other methodologies like cross-validation and model size considerations have received little attention.

B. Other Datasets and Methodological Roots

Other public benchmark datasets also reveal such behaviors. Gururaj et al. [17] show that careful preprocessing along with transfer learning can produce clinically reasonable results on HAM10000 dataset [18]. Suganthi et al. [19] use the stacking of InceptionV3 and MobileNetV2 as logistic regression meta-learner and reach the accuracy of 99.2%. Pandey et al. [20] follow the opposite approach—denoising through non-local means and sparse dictionary learning before standard CNN, achieving 85.61% on HAM10000 and 81.23% on ISIC-2019. SIIM-ISIC [21] also demonstrates similar tendencies.

Our current work is based on three existing ideas: feature generalization in transfer learning [22]; RepViT [6], that gives hierarchical features at mobile-scale computational budget; and DCT-based architecture in FcaNet [4], [5], that expands SE attention [3] outside the average pooling. Prior works on the MSCD-10K dataset do not use all three ideas.

III. Proposed Methodology

A. Dataset and Preprocessing

The MSCD-10K dataset [7], comprising 10,605 dermoscopic images, is used in this study. The development set contains 9,605 images (5,000 benign, 4,605 malignant), while the holdout test set contains 1,000 images (500 from each class). Fig. 1 shows representative examples; the visual overlap between classes underscores the task's difficulty.

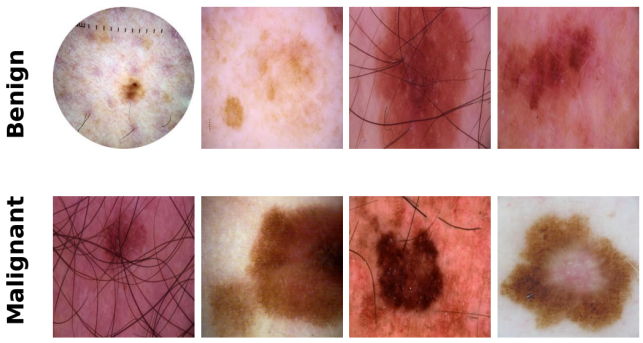


Fig. 1. Sample dermoscopic images from the MSCD-10K dataset [7]. Top row: benign lesions; bottom row: malignant lesions.

During training, images undergo random resized cropping (scale 0.8–1.0), horizontal and vertical flipping, rotation within $\pm 15°$, and mild color jittering. All images are resized to 224×224 and normalized using ImageNet statistics. To address mild class imbalance, a positive-class weight $w_+ = 1.0858$ is applied to the loss function. Fig. 2 illustrates the complete pipeline.

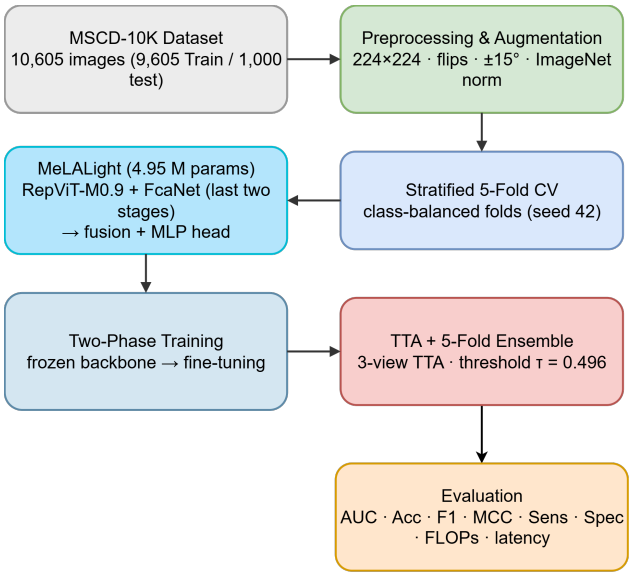


Fig. 2. Overall workflow of the proposed MeLALight methodology.

B. Backbone and Multi-Stage Attention

MeLALight uses a pretrained RepViT-M0.9 backbone [6] (Fig. 3) to extract hierarchical feature maps $\{F_0, F_1, F_2, F_3\}$ through a stem and four stages of progressively reduced resolution. We select the final two stages—$F_2 \in \mathbb{R}^{192\times14\times14}$ and $F_3 \in \mathbb{R}^{384\times7\times7}$—to balance spatial resolution and semantic context. Earlier stages remain in the backbone to condition deeper representations but are not used by the classification head. Fig. 4 shows the complete architecture.

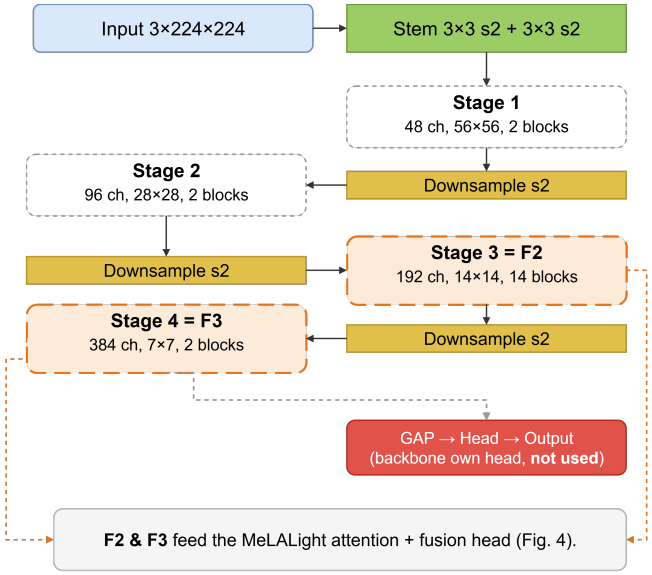


Fig. 3. RepViT-M0.9 backbone.

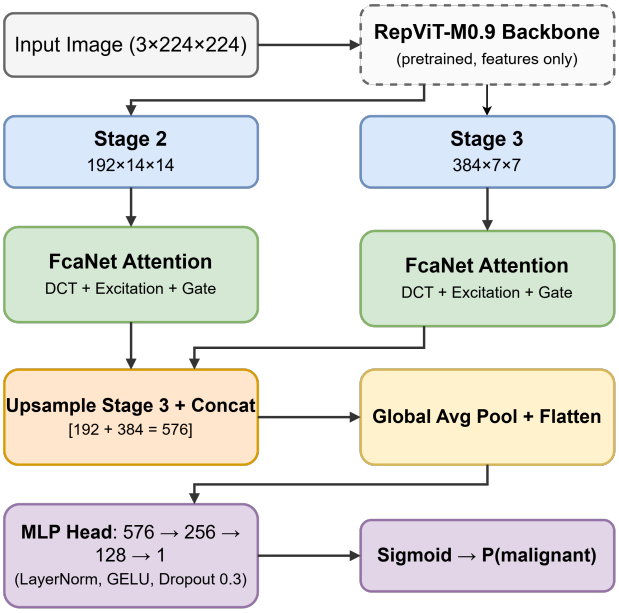


Fig. 4. Architecture of the proposed MeLALight model using frequency-channel attention on the last two RepViT stages.

Instead of global average pooling, we apply DCT-based FcaNet frequency-channel attention [4] independently to F_2 and F_3. The attention module (Fig. 5) pools each feature map against fixed DCT bases to construct a frequency descriptor, which then drives a compact excitation network. The 2D DCT basis functions are:

$$B_{u,v}(h,w) = \cos\left[\frac{\pi(2h+1)u}{2H}\right]\cos\left[\frac{\pi(2w+1)v}{2W}\right] \quad (1)$$

For a feature map X, the frequency descriptor for channel c is:

$$z_c = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} X_c(h,w)B_{u,v}(h,w) \quad (2)$$

The descriptor z passes through a reduction-8 excitation network:

$$s = \sigma(W_2\delta(W_1 z)) \quad (3)$$

where δ is ReLU and σ is sigmoid. The recalibrated features are $\tilde{X} = X \odot s$. Because the DCT bases are fixed, the two attention modules together add only ∼0.05M parameters.

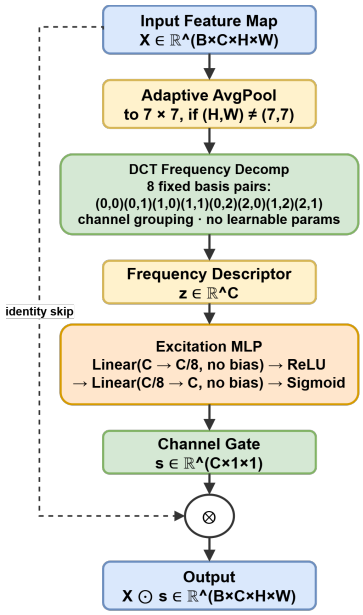


Fig. 5. DCT-based frequency-channel attention module. The module is channel-only, uses parameter-free DCT bases, and adds only two bias-free linear layers per attended stage.

C. Feature Fusion and Classification Head

The attended Stage-3 map is bilinearly upsampled to 14×14 and concatenated with the attended Stage-2 map:

$$F_{\text{fused}} = [\mathcal{A}(F_2) \,\|\, U(\mathcal{A}(F_3))] \quad (4)$$

This yields 576 channels. Global average pooling reduces this to a 576-dimensional vector, which feeds a lightweight head: LayerNorm → Linear(576, 256) → GELU → Dropout(0.3) → Linear(256, 128) → GELU → Dropout(0.3) → Linear(128, 1). Training minimizes binary cross-entropy with logits.

D. Training and Inference Protocol

Stratified five-fold cross-validation is used alongside a two-stage transfer learning approach. The first stage includes fixing the backbone (LR=10^{-4}) for adapting attention and head. The second stage consists of unfixing the backbone for fine-tuning (LR=10^{-5}). We optimize with the help of the Adam optimizer (weight decay = 10^{-5}) and scheduling methods such as ReduceLROnPlateau and early stopping. The early-stopping

patience counter is reset at the phase transition, giving the unfrozen backbone adequate time to converge.

The decision threshold τ is optimized per fold using validation F1-score maximization, and the final threshold is calculated using the average of the threshold per fold ($\tau_{\text{final}} = 0.496$). At inference, the 5-fold ensemble evaluates three views per image (original, horizontally flipped, and center-cropped), and the predicted probabilities are averaged before applying τ_{final}.

IV. Experiments and Results

A. Classification Performance

Table I shows the performance achieved through 5-fold cross-validation together with that of independent testing. The range of the AUC is quite narrow, i.e., between 0.9777 and 0.9850, whereas the range of the accuracy is between 93.28% and 94.01%. The minimal variance means that the performance does not change regardless of how the dataset is split into folds. Training was smooth in all the folds, and checkpoints with the best performance happened to be at epochs between 51 and 58 (Phase 2).

The performance on the hold-out set of 1,000 images is accuracy 91.80% and AUC 0.9772. The ROC curve (Fig. 6) illustrates a high value of true positives with respect to low values of false positives. All metrics mentioned use the same threshold $\tau = 0.496$.

TABLE I
Classification Performance

Metric	5-Fold CV (Mean ± Std)	Test Set
AUC	0.9817 ± 0.0026	0.9772
Accuracy	93.64% ± 0.35%	91.80%
Precision	95.21% ± 1.05%	95.24%
Recall	91.34% ± 0.61%	88.00%
Specificity	95.76% ± 1.00%	95.60%
F1-Score	93.23% ± 0.34%	91.48%
MCC	87.31% ± 0.73%	83.84%

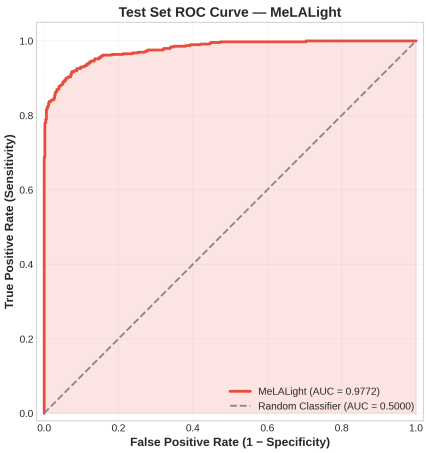


Fig. 6. Test set ROC curve of the five-fold ensemble (AUC = 0.9772).

B. Class-wise Behaviour and Confusion Matrix

The confusion matrix (Fig. 7) highlights both the model's capabilities and constraints. From 500 benign images, 478 are correctly identified with specificity of 95.60%, and 22 healthy people will be unnecessarily referred for biopsies. On the other hand, out of 500 melanoma cases, 440 are correctly identified, giving recall of 88.00% and missing 60 melanomas. In cases where the model classifies the image as malignant, it can be seen that the model is very reliable as evidenced by its precision of 95.24% and F1 score of 91.48%.

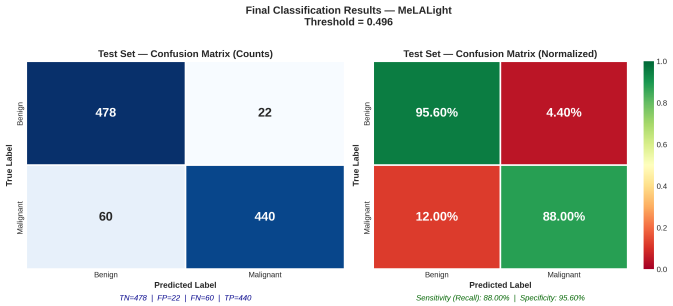


Fig. 7. Confusion matrices of MeLALight on the independent test set at threshold 0.496. Left: absolute counts (TN=478, FP=22, FN=60, TP=440); right: normalized percentages.

C. Computational Efficiency

The hardware cost with an input size of 224×224 is illustrated in Table II. The whole architecture contains 4.95 million parameters and 1.6335 billion FLOPs (0.8167 billion MACs). As shown by the parameter analysis, the RepViT backbone takes up 95.4% of the total model size, while the two FcaNet branches and the classifier share 0.9% and 3.7%, respectively. The application of frequency-domain attention does not bring much extra computation cost. The inference speed is 12.48 ms/image (batch size = 1) with a rate of 690.9 images/s on the reference GPU.

TABLE II
Computational Efficiency (224×224, FP32)

Metric	Value
Total Parameters	4.95 M
FLOPs	1.6335 B
MACs	0.8167 B
Latency (BS=1)	12.48 ms
Latency (BS=32)	46.31 ms
Peak GPU Memory	175.4 MB
Throughput	690.9 img/s

D. Ablation Study: Attention Placement

Comparison of different attention placements is shown in Table III. Attending to all four stages (configuration D) provides only slightly better AUC value by 0.0005 but increases memory consumption up to 651.4 MB. Attending to the last stage only (configuration A) is the most unstable one ($\sigma_{\text{AUC}} = 0.0064$) and causes a shift in threshold to 0.425. Two-stage configuration (configuration B) is the most stable

one ($\sigma_{\mathrm{AUC}} = 0.0026$), also it gives the best results in precision (95.24%) and specificity (95.60%) which is the most important criteria in screening.

TABLE III
ABLATION ON ATTENTION PLACEMENT. BEST VALUE PER COLUMN IN BOLD.

Config	AUC	Acc (%)	σ_{AUC}	Params (M)	Mem (MB)
A (1-stage)	0.9759	91.40	0.0064	4.89	175.2
B (2-stage)	**0.9772**	**91.80**	**0.0026**	**4.95**	**175.4**
C (3-stage)	0.9758	91.70	0.0037	4.97	217.6
D (4-stage)	0.9777	91.90	0.0032	4.99	651.4

E. Comparison with State-of-the-Art

Table IV compares **MeLALight** with some of the state-of-the-art methods on MSCD-10K [7]. This work is positioned as the second smallest model in terms of number of parameters in comparison with other methods. The main novelty of the suggested approach is the integration of multi-stage, DCT-based FcaNet frequency-channel attention on the last two RepViT backbone stages. Despite the fact that some models have better accuracy (Hamim et al., Balaha et al.), they are built using bigger backbones and do not give AUC. Moreover, some ensemble approaches (Albashish et al.) include more than 100 million parameters. The assumption is that a slight drop in accuracy is tolerable in case of usage on a cheap device with reproducible results.

TABLE IV
COMPARISON ON MSCD-10K. NR = NOT REPORTED; † PARAMS FROM PUBLIC SPECS; ‡ MSCD-10K + SIIM-ISIC; ^{a} ABSTRACT VALUE (BODY: $90.86\% \pm 0.42$).

Study	Architecture	Params (M)	Acc. (%)
Hamim et al. [1] (SmartSkin-XAI)	DenseNet-121 + Grad-CAM	8.2	98.00
Balaha et al. [16]	U-Net + DenseNet-169 + HHO	14.3†	97.08
Koteswara Rao et al. [10]	ResNet-50 + ViT	$\sim$112†	94.00
Al Huda et al. [13] (DeepMelaNet)	Modified EfficientNet	NR	93.40
Rao et al. [11]	CNN–LSTM	NR	93.40
Ghosh et al. [9]	DCNN + CapsNet + ViT ens.	NR	91.60
Albashish et al. [2]	VGG16 + NASNet + ResNet50V2	$\sim$169†	91.50^{a}
Quishpe-Usca et al. [14]	AlexNet + wavelet + hair rem.	61.1†	91.30
Yaqoob et al. [15]‡	VGG-16 transfer learning	138.4†	89.00
Sabir & Mehmood [8]	ResNet-18	11.7†	87.00
Proposed MeLALight	**RepViT-M0.9 + FcaNet**	**4.95**	**91.80**

V. CONCLUSION

In this work, we have considered how well a compact classifier for melanoma classification could perform when its feature recalibration happens in the frequency domain. The answer is: about as far as models several times its size. Coupling RepViT-M0.9 with DCT-based FcaNet frequency-channel attention on the last two stages delivers consistent cross-validation performance, 91.80% holdout accuracy, and under 5M parameters, and the ablations confirm that two attended stages are the right compromise. We explicitly position **MeLALight** as a first-pass triage tool for resource-constrained environments — for example, a mobile screening app in a rural clinic, where running a 169M-parameter ensemble is impossible — that clears obviously benign cases and routes suspicious lesions to a clinician.

Limitations of the study need to be considered in addition to the above results: conclusions are based solely on one public dataset; the task is binary, while clinical practice identifies several lesion types; only images have been used (patient's age, gender, lesion localization and other clinical information was not available); feature profiling was performed on GPU, not on the edge device. 60 out of 500 melanomas were not detected by the system, which is an acceptable rate for a second reader, but not sufficient for an autonomous system. Further development is expected to include: external and multi-center validation (e.g. HAM10000, ISIC), multi-class classification, inclusion of clinical metadata, Grad-CAM feature explanation and mobile deployment study of the 4.95M parameters model.

We believe that **MeLALight** represents a meaningful step toward practical, accessible AI-driven melanoma screening, demonstrating that high accuracy need not come at the cost of computational efficiency.

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